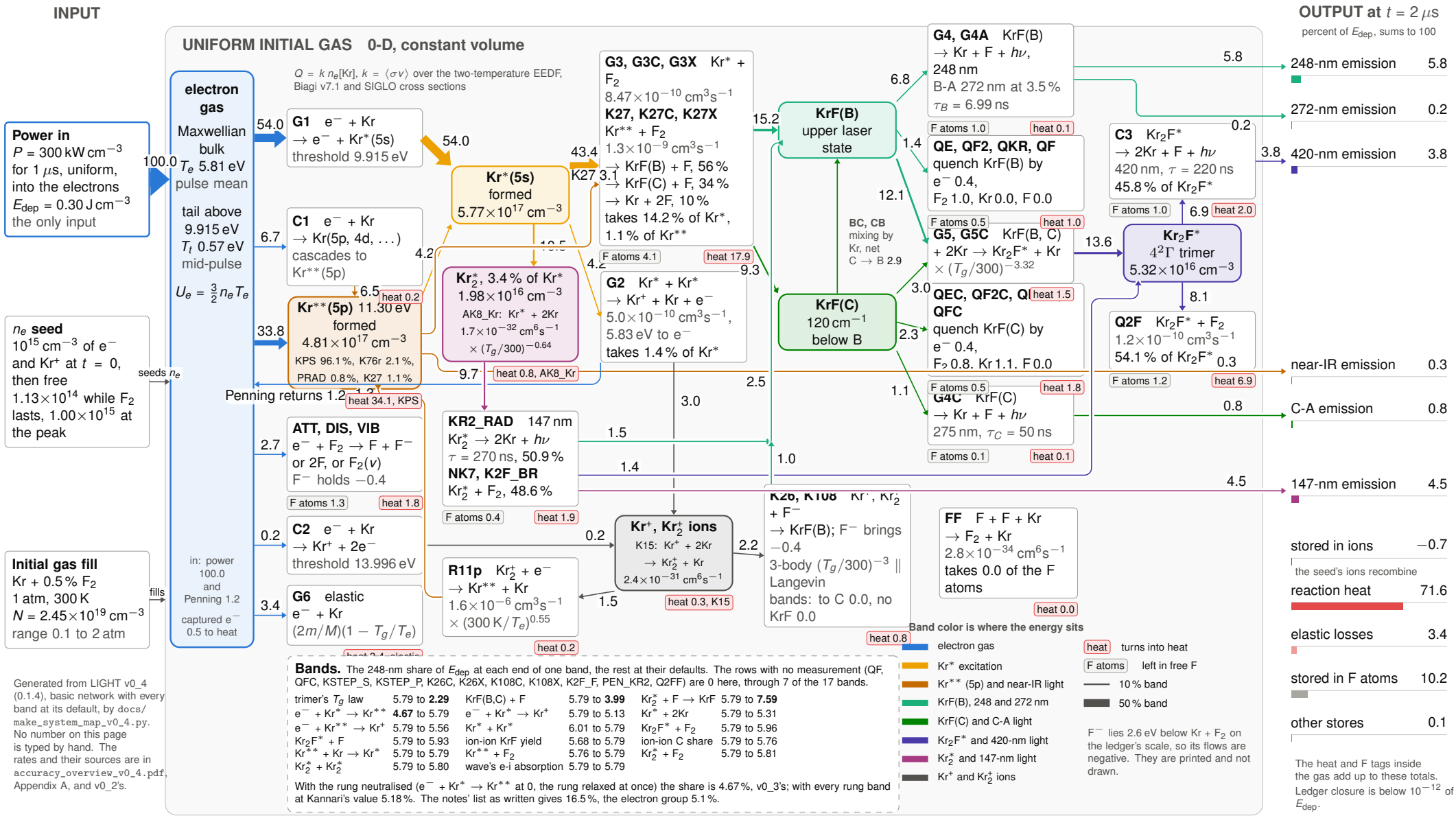
LIGHT system map, v0_4: energy flow through the Kr/F_2 kinetics with the 5p rung

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs on v0_2's electron closure. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



Bands. The 248-nm share of E_{dep} at each end of one band, the rest at their defaults. The rows with no measurement (QF, QFC, KSTEP_S, KSTEP_P, K26C, K26X, K108C, K108X, K2F_F, PEN_KR2, Q2FF) are 0 here, through 7 of the 17 bands.

trimer's T_g law	5.79 to **2.29**	KrF(B,C) + F	5.79 to **3.99**	Kr^*_2 + F → KrF	5.79 to **7.59**
e^- + Kr^* → Kr^{**}	**4.67** to 5.79	e^- + Kr^* → Kr^+	5.79 to 5.13	Kr^* + 2Kr	5.79 to 5.31
e^- + Kr^{**} → Kr^+	5.79 to 5.56	Kr^* + Kr^*	6.01 to 5.79	Kr_2F^* + F_2	5.79 to 5.96
Kr_2F^* + F	5.79 to 5.93	ion-ion KrF yield	5.68 to 5.79	ion-ion C share	5.79 to 5.76
Kr^{**} + Kr → Kr^*	5.79 to 5.79	Kr^{**} + F_2	5.76 to 5.79	Kr^*_2 + F_2	5.79 to 5.81
Kr^*_2 + Kr^*_2	5.79 to 5.80	wave's e-i absorption	5.79 to 5.79		

With the rung neutralised (e^- + Kr^* → Kr^{**} at 0, the rung relaxed at once) the share is 4.67 %, v0_3's; with every rung band at Kannari's value 5.18 %. The notes' list as written gives 16.5 %, the electron group 5.1 %.

Generated from LIGHT v0_4 (0.1.4), basic network with every band at its default, by `docs/make_system_map_v0_4.py`. No number on this page is typed by hand. The rates and their sources are in `accuracy_overview_v0_4.pdf`, Appendix A, and v0_2's.

The notes' outputs as flows, which overlap: to Kr^* 60.6 %, to Kr^+ 3.2 %, to KrF(B) and KrF(C) 23.0 %, heat 71.6 %, elastic losses 3.4 %, emission 15.5 %. **The 5p rung.** The 5s ↔ 5p electron cycle (K76, K76r) circulates 0.8 times the Kr^* formed, so it is drawn as its two net bands: of the Kr^* formed, 81.0 % is pumped up by the electrons (net of the return), the rung relaxes 96.1 % of it to Kr^* on Kr (KPS, 1.7×10^{-10} cm^{3}s^{-1}; PRAD, τ = 29 ns, radiates 0.8 %), and the gap of 1.385 eV per cycle takes 33.8 % of the input as heat. Of the KrF formed, 8.63×10^{16} cm^{-3}, 3.6 % came from ion-ion recombination, 5.6 % from Kr^*_2 + F_2 and 5.3 % from Kr^{**} + F_2; 25.1 % radiates at 248 nm, 56.1 % forms the trimer. At the end T_g = 742 K, p = 2.48 atm; F_2 fell below 10 % of the fill at 848 ns. **Optics at 248** nm (`optics.py`, a diagnostic, no cavity), over 500 to 1000 ns: g_0 = 4.12, α = 0.63, $g_0 - \alpha$ = 3.50 %/cm (g_0 peaks at 5.10 at 560 ns), $\sigma_B = 2.6\times10^{-16}$ cm^2, α by species Kr_2F^* 0.44, Kr^{**} 0.09, Kr^* 0.03, F_2 0.02, Kr^+_2 0.02, F^- 0.02 %/cm, Kr_2F^* at Hakuta et al.'s measured 1.0×10^{-18} cm^2.